Terafence
Reliable Transformations!

Large File Transfers

with complete galvanic network isolation



TFG121-BF



Technology Overview

Terafence is proud to introduce a unique, secure, and innovative network security appliance – Terafence TFG121-BF, a dynamic, configurable, agile, and complete unidirectional Secure Gateway solution including the FPGA data-diode.

Terafence manufactures Cyber Security hardware products to allow security architects the freedom in protecting their networks as needed.

Terafence TFG121-BF is a stand-alone unit that offers unique features and capabilities as a unidirectional, data-diode based device. Acting as a full Protocol Breaker or proxy server, total network segmentation, Air-Gap level, is guaranteed.

TFG121-BF utilizes a proprietary FPGA-based hardware architecture, designed, and manufactured by Terafence.

Certifications & Compliances:

- IEC62443-4-2 SL2
- FCC, EMI/EMC, CE, RoHS, MIL-STD-810F.
- IP30 protection, SECC, Black Metal chassis

Why choose TFG121-BF?

Key Features

- Complete galvanic network isolation for enhanced security and protection.
- Handles large file transfers of 250 GB+ single files.
- Complete all-in-one solution with no need for additional software or hardware.
- Proprietary Terafence FPGA-based hardware architecture for hardware-enforced security.
- Two dedicated CPUs for protocol support, processing, and termination.
- Secure HTTPS-based web GUI for configuration and management, accessible only from the secure side.
- Hardened operating system on the dedicated CPUs.
- Core security hardware operates independently without an operating system, MAC address, or IP address.

Supported Protocols

TFG121-BF supports multiple, simultaneous industrial and IT protocols, including:
SFTP/FTP/SMB – Secure file or folder-tree transfers (tested <250 GB+ files)
SMTP – Email relay
SYSLOG Forwarding – Log event transmission
RTSP/ONVIF – Live streaming for CCTV (50 CCTV max)
HTTP/S – File and data uploads to web servers or cloud platforms
Local SYSLOG Forwarding – Internal log event management
Modbus – Communication between multiple PLCs and HMIs
IEC-104 – Protocol for power utilities
MQTT – Data brokering and publishing
And various other protocols as per customer requirements.

Network Interfaces:	1 x Data Port, 1 x Management Port per side
Relative Humidity:	5% ~ 95% | 131°F, non-condensing
Environment:	Indoor use only
Cooling:	Dual fan cooling
Latency:	Protocol / data size dependent, core latency <2 nanoseconds

Technical Specifications

Product Type:	Unidirectional secure gateway / full protocol breaker with FPGA data diode – 19" rack version – customizable internal configuration
Data Throughput:	Up to 1 Gbps with auto-negotiation
Power Supply:	100–240 VAC, 50/60 Hz, with dual redundant internal hot-swappable AC power supplies
Form Factor:	1U rack-mountable appliance
Mounting Options:	19-inch 1U rack mount
Dimensions:	(W × H × D): 500 mm × 440.2 mm × 44.2 mm (1U)
Operating Temperature:	32°F to 113°F
Storage Temperature:	-40 °F to 185 °F
Processor:	Two dedicated CPUs, Intel i3 or better
RAM:	Customizable as per requirement
Local Storage:	Customizable as per requirement
Large-File Handling:	Dependent on SSD-M2 size, 250 GB tested
Reliability:	High availability, long-term operation, high MTBF (~10 years)

All images are for illustrative purposes only. Product configurations, specifications, and features may be customized to meet customer-specific requirements and may vary accordingly.

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